

Distributed Control and Communication Framework for Connectivity Preservation in Flow-Driven Robot Fleets

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Abstract—We consider deployment of underwater robot fleets in flow-dominated environments where communication is short-range and unreliable, localization may be denied, and actuation energy is limited. The key systems challenge is to maintain end-to-end connectivity under flow-driven dispersion while executing a mission objective, under realistic communication constraints.

We propose a two-layer architecture. First, a distributed topology-maintenance layer sparsifies the time-varying communication graph to a connected, spanning-tree-like subgraph using only local neighbor exchanges. In contrast to communication-heavy graph methods, the topology layer relies on constant-size messages and local parent selection to prune redundant edges; we explicitly account for bytes per message, messages per update, and delivered bandwidth, and evaluate robustness under packet drops, random delays, asynchronous broadcasts, and neighbor churn. Second, each robot solves a standard Controlled Lyapunov Functions (CLFs) and Controlled Barrier Functions (CBFs) with quadratic program (QP) that minimally modifies a nominal goal-seeking controller to satisfy collision-avoidance and range-maintenance constraints on the maintained edges.

We validate the complete pipeline in 3D HoloOcean simulations with multiple BlueROV2 vehicles under an advection-diffusion motion model, demonstrating goal-directed navigation while preserving end-to-end connectivity with measured communication budgets under impaired channels.

I. INTRODUCTION

We are interested in deploying a fleet of underwater robots into large-scale, challenging environments such as the deep ocean and under-ice cavities, where localization is denied, communication bandwidth is limited, and the environment is dominated by dynamic flows and eddies [1]. These robots must traverse long distances on limited energy budgets, discover emerging regions of interest, sample the environment, and relay data, all while relying on intermittent and bandwidth-limited communication [2], [3].

The central challenge is to design strategies that enable the fleet to efficiently cover a large spatial domain to discover emerging targets while maintaining network connectivity, and realize a globally connected configuration using only localized information and limited communication.

Our proposed approach is to deploy the robots together and then allow them to drift with the prevailing flow. Control effort is only applied when it is necessary. This includes

situations when the robots risk drifting outside a prescribed region or when they risk losing connectivity. In this way the formation disperses while staying connected, and the group can also reconfigure when a new target location is discovered or generated [4]. This strategy is bold, but it is supported by recent progress. Previous work has shown that surface vehicles can drift with the flow while using minimal control to remain in gyre like structures [5], or track the periodicity of revisits via intermittent control triggered by prescribed conditions [6]. In addition, broadly adopted models describe the uncontrolled motion of objects in water as drift combined with Brownian type diffusion [7]–[9], which align with observation on underwater robots [10], [11]. This indicates that our fleet will approximately follow the main direction of the flow while also dispersing stochastically.

Although dispersion promotes coverage of large task spaces, two challenges remain open: maintaining end-to-end connectivity under flow-driven neighbor churn, and enabling goal-driven reconfiguration using only localized information and unreliable, bandwidth-limited communication. Prior work on multi-robot connectivity has emphasized graph-theoretic formulations [12], [13] and control-barrier-function (CBF) approaches for energy-efficient constraint enforcement. A representative direction maintains connectivity through spectral quantities such as the Fiedler value [14], which can be effective but typically requires global information (or centralized computation) and is expensive to maintain online. Distributed CBF formulations avoid global spectral computations by enforcing local pairwise range constraints [15], [16]; however, applying constraints to all instantaneous neighbors can be communication- and control-intensive in dense graphs, especially when the neighbor set changes under dispersion.

To reduce this burden, recent hybrid frameworks restrict connectivity maintenance to a sparse coordination structure, e.g., using a minimum spanning tree (MST) and then applying local CBF constraints only on selected links [17]. This idea is promising for scalability, but constructing such sparse structures is classically communication-intensive in fully distributed settings: exact distributed MST protocols (e.g., GHS-style methods and variants) require substantial coordination and message complexity [18], [19], and practical alternatives trade optimality for reduced overhead [20]. For our setting, the key systems constraint is not optimality of the sparse topology but feasibility under underwater-style communication limits and impairments. Recent progress has shown that connectivity-relevant structure can be inferred

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using distributed procedures with limited information exchange [21]. Motivated by these observations, we construct a spanning-tree coordination structure using constant-size neighbor broadcasts and local parent selection, and we explicitly quantify bytes per message, messages per epoch, and delivered bandwidth under packet drops, random delays, asynchronous updates, and neighbor churn.

Upon the maintained spanning-tree structure, we apply a standard CLF–CBF–QP safety filter. **Systems contributions** (i) A communication-feasible distributed pruning protocol that constructs a connected spanning-tree topology with constant-size messages per broadcast and local keep/prune decisions, together with measured communication budgets. (ii) A closed-loop integration of topology maintenance with CLF–CBF control for goal-reaching under link-maintenance constraints on the maintained tree edges. (iii) A 3D simulation evaluation in HoloOcean with BlueROV2 vehicles under an advection–diffusion motion model, including practical disturbances.

This paper is organized as follows. Sec. II introduces the problem and modeling assumptions. Sec. III presents the distributed topology-maintenance protocol and the CLF–CBF control layer. Sec. IV describes the communication impairments and experimental setup and reports results from communication stress tests and 3D HoloOcean simulations. Sec. V concludes and outlines limitations.

II. BACKGROUND AND PROBLEM FORMULATION

We now formalize the problem. This section introduces the robot dynamics model, states the assumptions, defines the graph-theoretic notation, and presents the formal problem statement.

A. Advection-diffusion model [7]–[9]

We model a fleet of N robots operating in $\mathbb{R}^n$ (with $n = 2$, used for visualization in our simulations) released together in the water, their collective motion can be decomposed into a drift component induced by an ambient flow field and a stochastic dispersion component due to unexpected turbulence and small perturbations. Let $x_i(t) \in \mathbb{R}^n$ denote the position of the i -th robot at time t , where $i \in \{1, 2, \dots, N\}$, and let

$$\bar{x}(t) = \frac{1}{N} \sum_{i=1}^N x_i(t)$$

be the center of mass of the group.

The evolution of the center of mass is governed primarily by the advective flow field, i.e.,

$$\dot{\bar{x}}(t) \approx f_{\text{flow}}(\bar{x}(t), t),$$

where f_{flow} is the background velocity field. Each individual object then deviates from $\bar{x}(t)$ due to unresolved small-scale turbulence and fluctuations, which can be modeled as Brownian motion:

$$\begin{aligned} dx_i(t) &= (f_{\text{flow}}(x_i(t), t) + v_i(t))dt + \sigma_{\text{diff}}dW_i(t) \\ \dot{v}_i(t) &= u_i(t) - \gamma v_i(t) \end{aligned} \quad (1)$$

where $v_i(t) \in \mathbb{R}^n$ is the velocity of each individual robot, $u_i(t) \in \mathbb{R}^n$ is the thrust, $\gamma > 0$ is a velocity damping factor, and $W_i(t)$ is a Wiener process capturing diffusion (with intensity σ_{diff}).

Thus the group drifts together with the flow through the motion of $\bar{x}(t)$, while individuals spread stochastically around the drifting center due to the independent diffusion terms.

Discrete-time simulation In simulation we integrate (1) using Euler–Maruyama with step Δt :

$$\begin{aligned} x_i^{k+1} &= x_i^k + (f_{\text{flow}}(x_i^k, t_k) + v_i^k)\Delta t + \sigma_{\text{diff}}\sqrt{\Delta t}\xi_i^k, \\ \xi_i^k &\sim \mathcal{N}(0, I). \end{aligned} \quad (2)$$

together with the velocity update $v_i^{k+1} = v_i^k + (u_i^k - \gamma v_i^k)\Delta t$. This makes the diffusion term an explicit part of the state evolution used in our experimental evaluation.

We assume the ambient flow is spatially smooth and Lipschitz over the operating domain: $\|f_{\text{flow}}(x, t) - f_{\text{flow}}(y, t)\| \leq L_f\|x - y\|$. This captures non-uniform flows where relative drift generally does not cancel.

B. Control-affine model

For controller design we use a reduced control-affine representation of the position dynamics,

$$\dot{x}_i(t) = f_{\text{flow}}(x_i(t), t) + u_i(t) + d_i(t), \quad \|d_i(t)\| \leq \bar{d}, \quad (3)$$

where u_i is the commanded control input and d_i aggregates unmodeled effects into a bounded disturbance. We emphasize that (3) is used as a *design abstraction* for the CLF–CBF safety filter, while our simulations evolve robots under the advection–diffusion model in (2).

Because the flow is evaluated at each robot’s position, it does not cancel in relative coordinates when the flow is spatially varying. Defining $z_{ij} \triangleq x_i - x_j$, the relative dynamics satisfy

$$\dot{z}_{ij} = \Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - u_j) + (d_i - d_j), \quad (4)$$

with $\Delta f_{\text{flow}}(x_i, x_j, t) \triangleq f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$. Under intermittent communication, robot i does not have access to the instantaneous neighbor control $u_j(t)$. Following Assumption 3, we write

$$u_j(t) = \hat{u}_j(t) + e_{ij}(t), \quad \|e_{ij}(t)\| \leq \bar{e}, \quad (5)$$

where $\hat{u}_j(t)$ is the most recently received neighbor-control estimate and $e_{ij}(t)$ captures staleness. Substituting (5) into (4) yields the implementable form used in the robust CBF constraints:

$$\dot{z}_{ij} = \Delta f_{\text{flow}}(x_i, x_j, t) + (u_i - \hat{u}_j) + (d_i - d_j) - e_{ij}. \quad (6)$$

So, the flow contribution cancels only in the special case of spatially uniform flow. In our setting, spatial variation of f_{flow} is explicitly bounded, which is a quantity that enters the connectivity and invariance analysis.

The time derivative of the squared inter-robot distance can be written as

$$\dot{z}_{ij}^2 = 2z_{ij}^\top \left((f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)) + (v_i - v_j) \right)$$

This expression makes explicit that the flow contributes through the *difference term* $f_{\text{flow}}(x_i, t) - f_{\text{flow}}(x_j, t)$.

Notice that the choice of a two-dimensional environment is made solely for ease of visualization. The developed method is dimension-agnostic.

C. Assumptions on communication, control and connectivity

Assumption 1. At time t , the communication network is an undirected disk graph $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ with range $R_{\max} > 0$:

$$(i, j) \in \mathcal{E}_t \iff \|x_i(t) - x_j(t)\| \leq R_{\max}. \quad (7)$$

Each robot i communicates only with its neighbors in its neighboring set

$$\mathcal{N}_i(t) \triangleq \{j \in \mathcal{V} : \|x_i(t) - x_j(t)\| \leq R_{\max}\}, \quad (8)$$

using only neighbor communication without any central coordination. Each robot has a unique identifier to index neighbor messages consistently. The initial graph $\mathcal{G}_0$ is connected.

For each robot i and all $t \geq 0$, the control is bounded as $\|u_i(t)\| \leq u_{\max}$. The unmodeled disturbance is bounded by $\|d_i(t)\| \leq \bar{d}$. The ambient flow field $f_{\text{flow}}(x, t)$ is Lipschitz in space with constant L_f over the operating domain, i.e., $\|f_{\text{flow}}(x, t) - f_{\text{flow}}(y, t)\| \leq L_f \|x - y\|$ for all relevant x, y, t .

Assumption 2. There exist margins $\varepsilon_s > 0$ and $\varepsilon_c > 0$ such that the required inter-robot separation and communication constraints are simultaneously feasible:

$$0 < d_{\min} + \varepsilon_s < R_{\max} - \varepsilon_c.$$

Moreover, the available control authority is sufficient to counter worst-case relative disturbance over the communication range, i.e.,

$$u_{\max} > \bar{d} + L_f R_{\max}.$$

Assumption 3. For each maintained coordination link $(i, j) \in \mathcal{E}_{\text{topo}}(k)$, robot i receives from robot j a timestamped packet containing its most recently applied control. Let

$$\hat{u}_j(t) \triangleq u_j(t - \tau_{ij}(t))$$

denote the locally available estimate at robot i , where the staleness satisfies $\tau_{ij}(t) \leq \bar{\tau}$ for all t while (i, j) is maintained. Further assume a bounded control slew rate $\|\dot{u}_j(t)\| \leq \bar{\nu}$ for all j, t . Then the neighbor-control mismatch $e_{ij}(t) \triangleq u_j(t) - \hat{u}_j(t)$ is bounded as

$$\|e_{ij}(t)\| \leq \bar{e} \triangleq \bar{\nu} \bar{\tau}.$$

Communication impairments Communication is local and unreliable: packets may be dropped, delayed, and received asynchronously. We model this by (i) probabilistic packet drops, (ii) bounded random delays, and (iii) independent broadcast schedules per robot. Our topology-update protocol and experiments explicitly account for message size and update rate, and we report delivered bandwidth under these impairments in Sec. IV.

D. Problem Statement

Consider the multi-robot system introduced above with dynamics (1) and communication model (7)–(8). The robots are advected by a spatially varying ambient flow field $f_{\text{flow}}(x, t)$ and are subject to bounded disturbances $d_i(t)$, which together induce natural dispersion. While this dispersion promotes exploration, it can also stretch inter-robot distances and break communication links.

We focus on systems feasibility. The maintained subgraph should remain connected while reducing the number of maintained links and the required communication overhead, under the impaired-channel conditions described above.

Let $\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t)$ denote the physical disk graph induced by the range $R_{\max}$ (Assumption 1). During the mission, robots may intentionally sparsify the network by maintaining a time-varying coordination subgraph $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$ with $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$.

We formalize the hard safety/connectivity requirements using the constraint sets

$$\mathcal{C}_{\text{safe}} \triangleq \{x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \geq d_{\min}, \forall i \neq j\},$$

$$\mathcal{C}_{\text{conn}}(\mathcal{E}) \triangleq \{x \in (\mathbb{R}^n)^N : \|x_i - x_j\| \leq R_{\max}, \forall (i, j) \in \mathcal{E}\},$$

and the admissible input set $\mathcal{U} \triangleq \{(u_1, \dots, u_N) : \|u_i\| \leq u_{\max} \forall i\}$ (Assumption 1).

Targets are not known a priori and are detected only when an agent enters its sensing range. During exploration, the team should drift with f_{flow} while remaining connected and reducing communication redundancy. Upon target detection, agents should initiate goal-seeking behavior maintaining end-to-end connectivity across the fleet.

Each agent i has access only to local information: (i) its own state x_i ; (ii) relative states $\{x_j - x_i\}_{j \in \mathcal{N}_i(t)}$ and messages from one-hop neighbors; and (iii) local graph quantities, like two hop connection, are computable through neighbor exchanges. We denote this information set by $\mathcal{I}_i(t)$. No centralized computation, global maps, or spectral quantities are assumed; a spanning-tree coordination structure is constructed distributedly using one-hop neighbor messages. Distributed connectivity-preserving control with topology sparsification. Design distributed policies consisting of (i) a feedback control law $u_i(t) = \kappa_i(\mathcal{I}_i(t))$ for each agent and (ii) a distributed concurrent pruning rule that generates $\mathcal{E}_{\text{topo}}(t) \subseteq \mathcal{E}_t$, using only local information, such that for all $t \geq 0$:

- 1) $x(t) \in \mathcal{C}_{\text{safe}}$ and $x(t) \in \mathcal{C}_{\text{conn}}(\mathcal{E}_{\text{topo}}(t))$, and the maintained graph $\mathcal{G}_{\text{topo}}(t)$ remains connected.
- 2) The controller exploits advection and uses minimal actuation.
- 3) The team spreads for coverage during exploration and, while expansion figuring out the current graph structure from local knowledge and periodically constructing a spanning-tree coordination structure and pruning redundant edges using only local information to make a sparse graph network.

The proposed framework in Sec. III provides a fully distributed realization of these objectives via distributed topology sparsification and decentralized control.

III. METHODOLOGY

Our framework couples a *graph layer* with a *control layer*. At discrete update times t_k , the graph layer constructs a sparse connected coordination structure using only one-hop neighbor messages, and prunes all other links. Between update times, the control layer implements a CBF safety filter where a nominal CLF controller drives robots toward task goals, while CBF constraints enforce (i) collision avoidance with nearby robots and (ii) connectivity range maintenance for the maintained coordination links. Although the instantaneous disk graph $\mathcal{G}_t$ varies with advection and diffusion, the connectivity CBF constraints ensure that maintained links stay within range between updates and therefore do not drop from the disk graph.

Notation. Robots evolve under the control-affine model (3). The instantaneous communication disk graph is defined in (7)–(8). Let t_k denote the k -th topology update time. At update k , the graph layer outputs a maintained edge set $\mathcal{E}_{\text{topo}}(k) \subseteq \mathcal{E}_{t_k}$ that forms a spanning tree of $\mathcal{G}_{t_k}$. For robot i , define the *maintained neighbor set* $\mathcal{T}_i(k) \triangleq \{j : (i, j) \in \mathcal{E}_{\text{topo}}(k)\}$. Separately, define a *safety neighbor set* $\mathcal{S}_i(t)$ used only for collision avoidance.

A. Distributed topology maintenance

We maintain a sparse coordination topology $\mathcal{G}_{\text{topo}}(t) = (\mathcal{V}, \mathcal{E}_{\text{topo}}(t))$ as a connected subgraph of the instantaneous communication graph $\mathcal{G}(t) = (\mathcal{V}, \mathcal{E}(t))$, where $\mathcal{E}(t) = \{(i, j) : \|x_i(t) - x_j(t)\| \leq R_{\text{max}}\}$ is induced by the communication radius R_{max} . Topology updates are executed at discrete epochs; between epochs, the control layer enforces range constraints only on $\mathcal{E}_{\text{topo}}$.

Under underwater-style constraints, the topology layer must avoid exchanging full state per robot. We therefore restrict messages to constant-size broadcasts and explicitly account for bytes/message and delivered bandwidth.

1) *Scalar hop-to-root propagation*: At the start of each topology-update epoch, a designated root robot r initializes a scalar hop estimate $\delta_r = 0$, while all other robots initialize $\delta_i = \infty$. Each robot i periodically broadcasts the constant-size tuple (i, δ_i, s_i) to its current one-hop neighbors, where s_i is a local sequence counter. Upon receiving a message (j, δ_j, s_j) from neighbor j , robot i updates

$$\delta_i \leftarrow \min\{\delta_i, \delta_j + 1\}, \quad (9)$$

and records j as a candidate parent if the minimum is achieved. This is an online unweighted Bellman Ford Search (BFS) update over the time-varying neighbor graph.

2) *Local parent selection and pruning*: After a fixed epoch duration T_{prune} , each robot $i \neq r$ selects a parent

$$p(i) \in \arg \min_{j \in \mathcal{N}_i(t)} \delta_j \quad \text{s.t.} \quad \delta_{p(i)} = \delta_i - 1, \quad (10)$$

Ties are broken by heuristics (e.g. smallest ID). The maintained edge set is then defined by the local keep rule:

$$\mathcal{E}_{\text{topo}} \triangleq \{(i, p(i)) : i \neq r, p(i) \text{ defined}\}. \quad (11)$$

Thus each robot maintains at most one outgoing topology edge; collectively, $\mathcal{G}_{\text{topo}}$ is a spanning-tree-like coordination structure whenever the epoch communication graph is connected and messages propagate.

a) *Communication accounting*: Each broadcast contains constant-size fields (sender ID, δ_i , and sequence counter). We report (i) bytes per message, (ii) messages per topology-update epoch, (iii) attempted (TX) and delivered (RX) bandwidth under packet drops and delays, and (iv) convergence time of the δ -propagation procedure.

b) *Robustness to asynchrony and packet impairments*: Robots broadcast asynchronously with jittered periods, and packets may be dropped or delayed. The implementation suppresses duplicates via sequence counters and terminates epochs based on observed stabilization. We quantify success rates (tree formation and root reachability) under these impairments in Sec. IV.

B. Illustrative example of a sparse graph

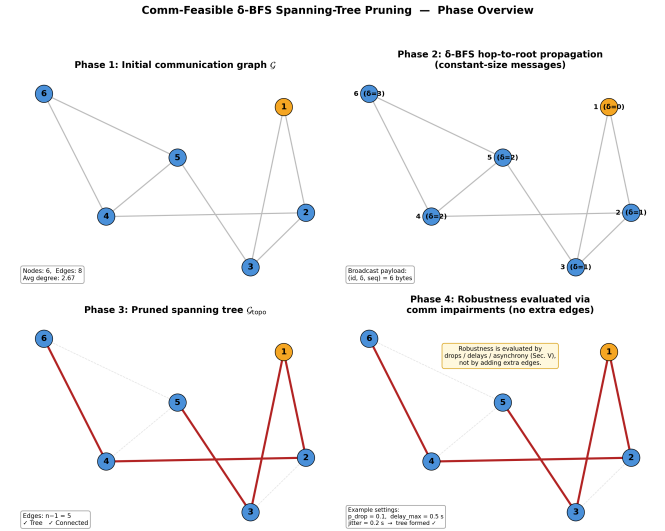


Fig. 1: Comm-feasible δ -BFS spanning-tree pruning on a 6-node example. **Phase 1**: initial communication graph $\mathcal{G}$. **Phase 2**: hop-to-root estimates δ_i propagate using constant-size neighbor broadcasts. **Phase 3**: local parent selection yields the pruned spanning tree $\mathcal{G}_{\text{topo}}$ with $n - 1$ maintained edges. **Phase 4**: robustness is evaluated via drops/delays/asynchrony (Sec. IV), *not* by adding extra edges.

In Fig. 1 we consider an undirected graph $G = (V, E)$ with $V = \{1, 2, 3, 4, 5, 6\}$, root $r = 1$, and

$$\begin{aligned} E = \{ & (1, 2), (1, 3), (2, 3), (2, 4), \\ & (3, 5), (4, 5), (4, 6), (5, 6) \}, \\ & |V| = 6, \quad |E| = 8. \end{aligned}$$

The neighbor sets are

$$\begin{aligned} N_1 &= \{2, 3\}, & N_2 &= \{1, 3, 4\}, & N_3 &= \{1, 2, 5\}, \\ N_4 &= \{2, 5, 6\}, & N_5 &= \{3, 4, 6\}, & N_6 &= \{4, 5\}. \end{aligned}$$

a) Phase 2: comm-feasible hop-to-root propagation.:

Rather than computing all-pairs hop distances, our pruning layer only maintains the *scalar* hop distance-to-root δ_i at each node. The root initializes $\delta_1 = 0$ and all other nodes initialize $\delta_i = \infty$. Robots periodically broadcast constant-size messages of the form

$$m_i = (\text{id} = i, \delta_i, \text{seq}_i),$$

and upon receiving m_j from neighbor j , node i applies the local update

$$\delta_i \leftarrow \min\{\delta_i, \delta_j + 1\}. \quad (12)$$

For this example, the converged hop-to-root distances are

$$\delta_1 = 0, \delta_2 = 1, \delta_3 = 1, \delta_4 = 2, \delta_5 = 2, \delta_6 = 3,$$

as shown in Phase 2 of Fig. 1.

b) Phase 3: local parent selection and pruning.: Each node selects its parent from neighbors that are exactly one hop closer to the root:

$$C_i = \{j \in N_i : \delta_j = \delta_i - 1\}, \quad p(i) = \min C_i.$$

This yields the parent pointers

$$p(2) = 1, \quad p(3) = 1, \quad p(4) = 2, \quad p(5) = 3, \quad p(6) = 4,$$

and the maintained topology is the set of parent edges

$$\begin{aligned} E_{\text{topo}} &= \{(1, 2), (1, 3), (2, 4), (3, 5), (4, 6)\}, \\ |E_{\text{topo}}| &= 5 = n - 1. \end{aligned}$$

Equivalently, for each $\{i, j\} \in E$ we keep the edge iff $p(i) = j$ or $p(j) = i$. The result is a spanning-tree coordination structure used by the control layer to enforce range-maintenance constraints only on the maintained edges.

c) Phase 4: robustness via impaired communication.:

In this paper robustness is evaluated empirically by injecting packet drops, random delays, and asynchronous broadcast schedules into the message passing and measuring (i) tree formation and root reachability, (ii) convergence time, and (iii) delivered bandwidth (Sec. IV). This keeps the topology layer consistent with underwater-style communication constraints.

This section describes the control layer used after a topology-update epoch produces the maintained edge set $\mathcal{E}_{\text{topo}}$ (Sec. III-A). We emphasize a systems viewpoint: the CLF-CBF quadratic program (QP) is used as a standard safety filter that enforces collision avoidance and range maintenance *only* on the maintained links, while allowing a nominal goal-seeking behavior. The novelty of this paper is not the CLF-CBF construction itself; rather, it is the communication-feasible topology layer and its robustness/communication evaluation under impaired channels.

Algorithm 1 Comm-feasible δ -BFS pruning (topology epoch)

State at robot i : $\delta_i \in \mathbb{N} \cup \{\infty\}$, parent $p(i)$, seq s_i , last-seq s_j^{last} for $j \in N_i$
Msg: (i, δ_i, s_i)
Init: if $i = r$ set $\delta_i \leftarrow 0$ else $\delta_i \leftarrow \infty$; $p(i) \leftarrow \emptyset$, $s_i \leftarrow 0$
while $t < T_{\text{prune}}$ and not stable for T_{stable} **do**
 periodically broadcast (i, δ_i, s_i) to neighbors
 if recv (j, δ_j, s_j) and $s_j > s_j^{\text{last}}$ **then**
 $s_j^{\text{last}} \leftarrow s_j$; $\delta_i \leftarrow \min(\delta_i, \delta_j + 1)$
 end if
end while
if $i \neq r$ and $\delta_i < \infty$ **then**
 pick $p(i) \in \{j \in N_i : \delta_j = \delta_i - 1\}$ (tie-break min ID)
end if
Output maintained edges $\mathcal{E}_{\text{topo}} = \{(i, p(i))\}$.

C. Nominal goal-seeking controller

Let $g \in \mathbb{R}^3$ denote a goal location. We define the Lyapunov goal-tracking function.

$$V_i(x_i) \triangleq \|x_i - g\|^2, \quad (13)$$

and use a nominal controller u_i^{nom} that drives V_i downward. In our implementation, u_i^{nom} is generated from a goal-directed velocity command and then mapped to vehicle actuation (Sec. IV-C). The CLF constraint can be written in the standard form

$$\dot{V}_i(x_i) \leq -c_V V_i(x_i) + \epsilon_i, \quad (14)$$

where $c_V > 0$ and $\epsilon_i \geq 0$ is a relaxation variable to guarantee feasibility when safety constraints dominate.

D. Safety constraints (CBFs) on maintained edges

We enforce safety/communication constraints only on the maintained topology edges $(i, j) \in \mathcal{E}_{\text{topo}}$. This is critical for scalability and communication feasibility: the control layer does not attempt to maintain all pairwise links in the instantaneous dense neighbor graph.

a) Range maintenance on maintained links.: For each maintained edge $(i, j) \in \mathcal{E}_{\text{topo}}$, define the range-maintenance barrier function

$$h_{ij}(x) \triangleq R_{\text{max}}^2 - \|x_i - x_j\|^2, \quad (15)$$

so that $h_{ij}(x) \geq 0$ corresponds to $\|x_i - x_j\| \leq R_{\text{max}}$. Using the relative dynamics in Sec. II-B, a standard CBF condition is

$$\dot{h}_{ij}(x) + \alpha_h h_{ij}(x) \geq 0, \quad \forall (i, j) \in \mathcal{E}_{\text{topo}}, \quad (16)$$

with class- $\mathcal{K}$ gain $\alpha_h > 0$.

b) Collision avoidance.: For collision avoidance, we require $\|x_i - x_j\| \geq d_{\text{min}}$ for nearby robots. A convenient barrier function is

$$h_{ij}^{\text{col}}(x) \triangleq \|x_i - x_j\|^2 - d_{\text{min}}^2, \quad (17)$$

with the CBF condition $\dot{h}_{ij}^{\text{col}}(x) + \alpha_c h_{ij}^{\text{col}}(x) \geq 0$ applied for pairs within a sensing/avoidance radius (this set can be chosen independently of $\mathcal{E}_{\text{topo}}$).

E. systems implementation

At each control update, robot i solves a small QP that minimally modifies the nominal command to satisfy the CBF constraints on its incident maintained edges:

$$\begin{aligned} u_i^* \in \arg \min_{u_i, \epsilon_i} \quad & \|u_i - u_i^{\text{nom}}\|^2 + \rho \epsilon_i^2 \\ \text{s.t.} \quad & \dot{V}_i(x_i) \leq -c_V V_i(x_i) + \epsilon_i, \\ & \dot{h}_{ij}(x) + \alpha_h h_{ij}(x) \geq 0, \quad \forall j : (i, j) \in \mathcal{E}_{\text{topo}}, \\ & \dot{h}_{ij}^{\text{col}}(x) + \alpha_c h_{ij}^{\text{col}}(x) \geq 0, \quad \forall j \in \mathcal{N}_i^{\text{col}}, \\ & \|u_i\| \leq u_{\max}, \quad \epsilon_i \geq 0, \end{aligned} \quad (18)$$

where $\rho > 0$ weights CLF relaxation. This formulation is standard in the CLF–CBF literature; we use it as a practical safety filter on the maintained topology.

a) Consistency with advection–diffusion simulation:

The QP is derived using the design model in Sec. II-B, while the experimental evaluation in Sec. IV evolves the robots under the advection–diffusion model (Sec. II-A) with packet drops/delays and neighbor churn. Thus the results quantify closed-loop robustness under the physics and communication impairments of interest, rather than relying on idealized synchronization assumptions.

IV. SIMULATIONS AND RESULTS

A. Simulation setup

We evaluate the proposed comm-feasible pruning layer (Alg. 1) and CLF–CBF safety filter in two complementary environments:

a) Synthetic stress tests (communication and pruning).:

To isolate communication feasibility and pruning robustness, we simulate the message-passing protocol over randomly generated connected graphs and inject packet drops, random delays, and asynchronous broadcasts. Each robot broadcasts a constant-size tuple $(\text{id}, \delta, \text{seq})$ and updates δ via (9), followed by local parent selection and pruning to $\mathcal{E}_{\text{topo}} = \{(i, p(i))\}$. These stress tests produce measured TX/RX message counts, delivered bandwidth, and convergence times under impaired channels.

b) 3D HoloOcean end-to-end simulations (advection–diffusion): To demonstrate realism, we implement the full pipeline in the 3D HoloOcean simulator with multiple BlueROV2 vehicles. Robot motion evolves under an advection–diffusion model (Sec. II) integrated via Discrete-time simulation (2) with spatially varying currents and diffusion, inducing neighbor churn in the disk graph (7)–(8). Topology updates run asynchronously during motion, and the maintained tree edges are used by the control layer to enforce range maintenance while executing a goal-reaching task.

c) Baselines.: We compare four maintained-topology strategies when applicable: **Fully Connected** (maintain all edges), **Centralized MST** (computed offline for comparison), **Adjacency Consensus** (sequential pruning baseline), and **Distributed Pruning (ours)**.

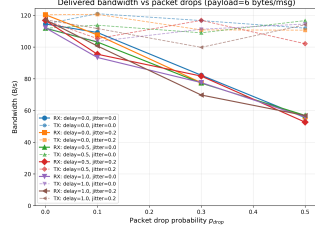


Fig. 2: Delivered bandwidth (RX) vs. packet drops under delay impairments.

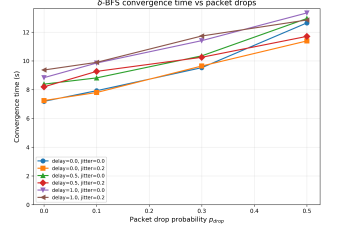


Fig. 3: δ -BFS convergence time vs. packet drops under delay impairments.

TABLE I: Communication budget (payload fixed at 6 bytes/msg)

N	Setting	Impairments	TX	RX	RX (B/s)	Conv. (s)
10	Ideal	$p=0, \tau=0$	135	135	114.6	7.19
		$j=0$				
10	Lossy	$p=0.1, \tau=0.5$	160	145	95.5	9.27
		$j=0.2$				
20	Ideal	$p=0, \tau=0$	354	354	274.0	7.86
		$j=0$				
20	Lossy	$p=0.1, \tau=0.5$	435	393	239.4	9.85
		$j=0.2$				

B. Communication accounting and stress tests

a) Message format and accounting.: Each pruning broadcast contains a constant-size tuple $(\text{id}, \delta, \text{seq})$ (payload = 6 bytes/message in our implementation, excluding transport overhead). We report both attempted transmissions (TX) and delivered receptions (RX). Bandwidth is computed as delivered payload bytes divided by the topology-update epoch duration; in our implementation epochs may end early upon stabilization (no δ updates for T_{stable}).

b) Stress-test protocol.: We sweep packet drops (p_{drop}), bounded random delays $\tau \sim \text{Unif}(0, \tau_{\text{max}})$, and asynchronous broadcast jitter. For each setting we measure: (i) tree formation success (connected and root-reachable), (ii) δ -BFS convergence time, and (iii) TX/RX message counts and bandwidth.

As shown in Fig. 2, delivered bandwidth decreases as packet drops increase and under larger delay/jitter. Fig. 3 shows that δ -BFS convergence time increases with impairments but remains within a short epoch horizon in our tested settings. Table I reports representative TX/RX message counts, delivered bandwidth, and convergence times for $N \in \{10, 20\}$.

C. Metrics

We report both topology-level and task-level metrics.

a) Topology formation and robustness.: A topology-update epoch is considered successful if the maintained graph $\mathcal{G}_{\text{topo}}$ is (i) connected and (ii) all nodes are reachable from the root (equivalently, $|\mathcal{E}_{\text{topo}}| = N - 1$ and $\mathcal{G}_{\text{topo}}$ is connected). We report:

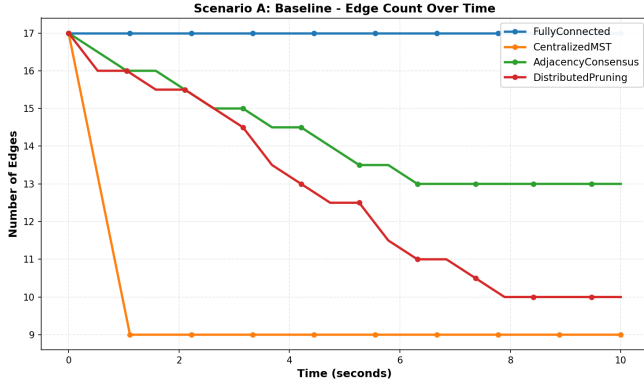


Fig. 4: Maintained edge count over time for four topology strategies (representative run).

- **Tree success rate (%)** over Monte Carlo trials under a given $(p_{\text{drop}}, \tau_{\text{max}}, \text{jitter})$.
- **Convergence time (s)**: time until δ values stabilize (no updates for T_{stable}) or the epoch ends at T_{prune} .
- **Communication cost**: TX/RX message counts per epoch and delivered bandwidth (B/s), as in Sec. IV-B.

b) Sparsification: We report the number of maintained edges $|\mathcal{E}_{\text{topo}}|$ and the edge reduction relative to the instantaneous disk graph:

$$\text{Edge reduction} = \frac{|\mathcal{E}(t_k)| - |\mathcal{E}_{\text{topo}}(k)|}{|\mathcal{E}(t_k)|} \times 100\%,$$

where $|\mathcal{E}_{\text{topo}}(k)| = N - 1$ by construction when the tree forms successfully.

c) Closed-loop task performance: For 3D HoloOcean runs we additionally report:

- **Goal-reaching time (s)** or final distance-to-goal at horizon.
- **Connectivity preservation**: fraction of time the maintained topology remains connected during motion.
- **Safety**: minimum inter-robot distance (collision avoidance).

D. Baseline comparison in lightweight simulation

We compare maintained-topology strategies in a lightweight simulator to illustrate sparsification dynamics and to provide a method-to-method baseline. This study is used for comparison under controlled conditions; realistic end-to-end validation in 3D HoloOcean under advection–diffusion currents and impaired communication is reported in Sec. IV-E.

a) Edge-count evolution: Fig. 4 shows the maintained edge count $|\mathcal{E}_{\text{topo}}(t)|$ over time for a representative run. The fully connected strategy maintains all links, while the centralized MST oracle immediately yields a tree with $N - 1$ edges. Our distributed pruning progressively reduces the maintained edge set toward a sparse, near-tree structure, whereas the sequential pruning baseline remains denser due to conservative removals.

E. 3D HoloOcean validation under advection–diffusion and impaired communication

We validate end-to-end operation in the 3D HoloOcean simulator with multiple BlueROV2 vehicles under flow-driven dispersion. Robot motion evolves under the advection–diffusion model (Sec. II) integrated via (2), which induces neighbor churn in the disk communication graph (7)–(8). During motion, robots execute topology-update epochs using the comm-feasible δ -BFS pruning protocol (Alg. 1) under packet drops, bounded random delays, and asynchronous broadcasts. The control layer then enforces range-maintenance constraints only on the maintained edges while robots pursue a goal.

Fig. 5 shows two camera views from a representative HoloOcean run, illustrating goal-directed deployment and the dense instantaneous communication links formed within range R_{max} . Fig. 6 reports the corresponding topology-update epoch: the instantaneous disk-graph communication network is pruned to a spanning tree ($|\mathcal{E}_{\text{topo}}| = N - 1$) while remaining connected, and the overlay reports TX/RX message counts and observed drops for that epoch. Together with the stress-test sweeps in Sec. IV-B, these results demonstrate systems feasibility under both realistic 3D dynamics and impaired communication.

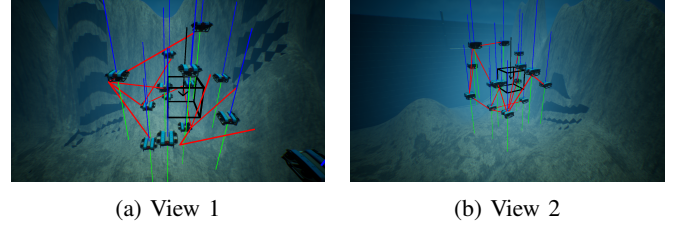


Fig. 5: 3D HoloOcean scene with multiple BlueROV2 vehicles during deployment under flow-driven dispersion. Red segments indicate pruned communication links within range R_{max} .

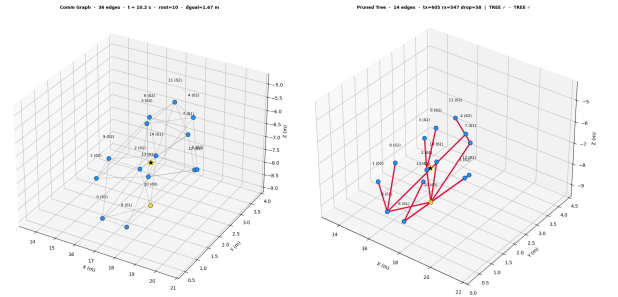


Fig. 6: HoloOcean pruning epoch ($N = 15$, $R_{\text{max}} = 2.0$ m) under impaired communication ($p_{\text{drop}} = 0.1$). Left: instantaneous disk-graph communication edges. Right: pruned maintained topology $\mathcal{G}_{\text{topo}}$ produced by the online δ -BFS protocol (Alg. 1), yielding a connected spanning tree. The overlay reports TX/RX counts and observed drops for the epoch.

V. CONCLUSION

We presented a systems-feasible, fully distributed framework for connectivity-preserving deployment of flow-driven robot fleets under bandwidth-limited and unreliable communication. The topology layer constructs a spanning-tree coordination structure using constant-size neighbor broadcasts and local parent selection and we explicitly quantify message budgets and delivered bandwidth under packet drops, random delays, and asynchronous broadcasts. On top of the maintained topology, each robot applies a standard CLF-CBF-QP safety filter that enforces collision avoidance and communication-range maintenance only on the maintained edges while executing goal-directed behavior under flow-driven dispersion.

We validated the approach through communication stress tests and 3D HoloOcean simulations with multiple BlueROV2 vehicles under an advection-diffusion motion model, demonstrating successful tree formation and end-to-end connectivity with measured communication budgets.

Future work will (i) incorporate higher-fidelity acoustic modem and packet-level network models, (ii) scale to larger teams and longer-duration missions with repeated reconfiguration events, and (iii) validate the framework on field robotic platforms in real currents.

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